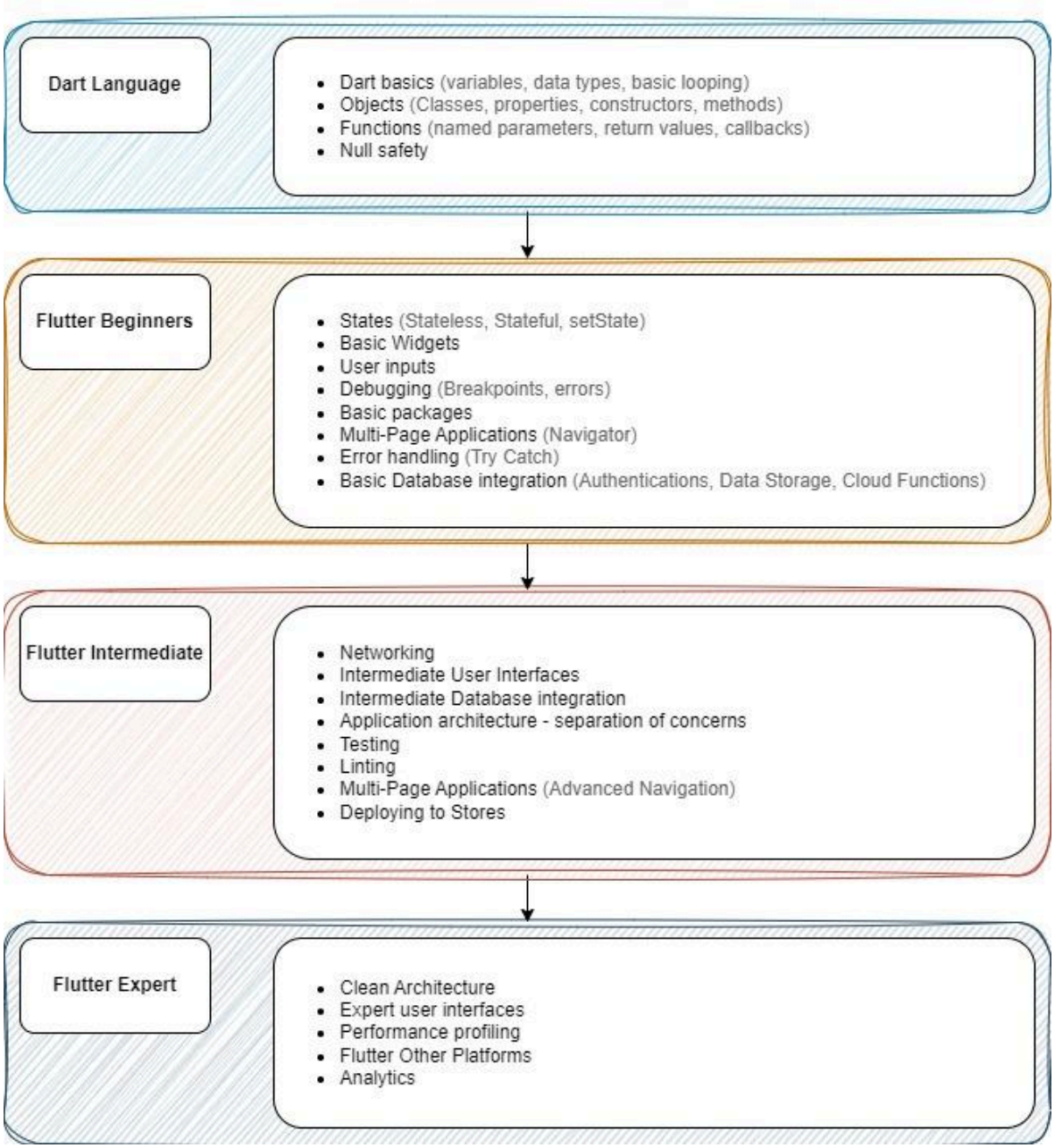
Flutter Roadmap

By Flutter Mapp

Overview:

- Diagram available for download also (Same email)



Dart Language

Dart basics:
- Variables
- Data types
    - Numbers
    - Booleans
    - Strings
    - Lists
    - Maps
- Basic looping
- Data fundamentals video (FREE): https://courses.fluttermapp.com/courses/flutter-for-beginners/lectures/43156060

Objects:
- Classes
- Properties
- Constructors
- Callbacks

Functions:
- Named parameters
- Return values
- Callbacks

Null Safety:
- Video (FREE): https://courses.fluttermapp.com/courses/how-to-learn-flutter-for-beginners/lectures/43158089

Flutter Beginners

States:
- Stateless
- Stateful
- setState

Basic Widgets:

- Padding - Expanded - StatelessWidget - StatefulWidget - MaterialApp - TextField - Wrap - Center - Scaffold - Appbar - Container - Image (Asset or Network) - Column - ElevatedButton - SizedBox - SingleChildScrollView - Icon(Icons.profile) - Colors.blue - List.generate - Row	- IconButton - MaterialBanner - TextButton - OutlinedButton - ListTile - ListView - Divider - TextStyle - Text - Visibility - TextFormField - SnackBar - StreamBuilder - BottomNavigationBar - Drawer - DrawerHeader - SafeArea - CircleAvatar - CheckBoxListTile - GestureDetector

User Inputs:
- TextField
- TextFormField
- Form

Debugging:
- Breakpoints
- Solving errors

Basic packages:
- Icon launcher: https://pub.dev/packages/flutter_launcher_icons
- Firebase core: https://pub.dev/packages/firebase_core
- Firebase auth: https://pub.dev/packages/firebase_auth

- Cloud firestore: https://pub.dev/packages/cloud_firestore
- SharedPreferences: https://pub.dev/packages/shared_preferences
- Other packages: https://pub.dev/

Multi-page application:
- Navigator 1.0
- Example:

```
onTap: () {
  Navigator.of(context).push(
    MaterialPageRoute(
      builder: (context) {
        return const HomePage();
      },
    ), // MaterialPageRoute
  );
},
```

Error handling:
- Try Catch

Basic Database integration:
- Authentications: https://pub.dev/packages/firebase_auth
    - Example: https://www.youtube.com/watch?v=rWamixHIKmQ
- Data Storage (cloud firestore): https://pub.dev/packages/cloud_firestore
- Cloud functions: https://pub.dev/packages/cloud_functions

Flutter Intermediate

Networking:
- Asynchronous Dart programming (Future, async, await)
- API Calls
- Json parsing
- Http packages (Dio): https://pub.dev/packages/dio

Intermediate user interfaces:
- Responsive UI
    - Example here: https://www.youtube.com/watch?v=yIxEpbyNAy8
    - Widget: LayoutBuilder
- Animations
    - Example here: https://www.youtube.com/watch?v=kJQCLrExrns

Intermediate Database integration:
- Google sign in
- Apple sign in
- Facebook sign in
- Relational database SQLite: https://pub.dev/packages/sqflite

Application architecture - separation of concerns:
- Riverpod (Flutter Mapp's favorite): https://pub.dev/packages/riverpod
- State managements
    - Provider
    - Riverpod
    - setState
    - InheritedWidget
    - Redux
    - BLoC
    - GetIt
    - MobX

Testing:
- Unit testing (FREE): https://courses.fluttermapp.com/courses/hero-to-pro/lectures/38288299
- Widget Testing (FREE): https://courses.fluttermapp.com/courses/hero-to-pro/lectures/38601758
- Integration Testing (FREE): https://courses.fluttermapp.com/courses/hero-to-pro/lectures/38613492
- Mocks

Linting:
- https://pub.dev/packages/lint

Multi-Page Applications:

- Navigator 2.0

Deploying to Stores:

- App Store
- Play Store

Flutter Expert:

Clean Architecture:

- Explained inside the Flutter Pro Course

Expert user interfaces:

- Widget lifecycle (Explained inside the Flutter Pro Course)
- Internationalizing Flutter apps: https://docs.flutter.dev/development/accessibility-and-localization/internationalization

Performance profiling:

- Explained inside the Flutter Pro Course
    - Performance overlay
    - Flutter Run Profile
    - Track Widget Builds
    - Network
    - Cpu profiler
    - Memory

Flutter Other Platforms:

- Flutter Web
- Flutter desktop

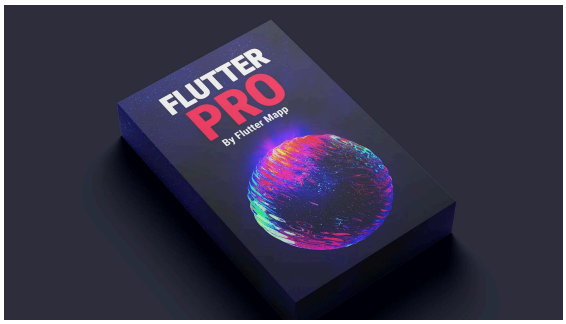
Analytics:

- Usage analytics
- Crash logging
- A/B testing

Courses:

Learn Flutter in 15 minutes (FREE)

- Video: https://youtu.be/aG9uSXdU2sk

Flutter For Beginners (Available on https://www.FlutterMapp.com):
Use code (40% OFF Limited Time Only) : **ROADMAP40**
Or use this link: Flutter For Beginners 40% OFF

Flutter Pro (Available on https://www.FlutterMapp.com):
Use code (40% OFF Limited Time Only): **ROADMAP40**
Or use this link: Flutter Pro 40% OFF